

Audit 10: current mathematics

Abstract

This packet studies how much order information survives in the normalized overlaps of principal upsets of a finite poset. A verified 16-element pair disproves universal reconstruction from the normalized Gram matrix G alone. What survives is exact recovery with labeled upset sizes, structural information readable from G , complete candidate-validity tests, and complete realization enumeration for positive-entry G . For the known collision, comparability-degree data distinguish the two members of its entire realization fiber. Whether G together with the unordered comparability-degree multiset determines every finite poset remains unresolved in this project.

Read this first

This is the current mathematical document at frozen checkpoint v130. It replaces the 316-page cumulative document as the reader entry point. The cumulative PDF and complete archive are preserved under 90_HISTORICAL_EVIDENCE. The historical record contains refuted targets and corrected routes; its dated claims are not all current conclusions.

What the data mean

For an element x , its principal upset U_x consists of x and all elements above x . Write $Z_{xy} = 1[x \leq y]$, $d_x = |U_x|$, $K = ZZ^T$, and $D = \text{diag}(d)$. Then $G_{xy} = |U_x \cap U_y| / \sqrt{d_x d_y}$, and $G = D^{-1/2} K D^{-1/2}$. We use exact $q_{xy} = G_{xy}^2$ in computations. A comparability degree counts the other elements comparable to x ; its multiset is supplied without a row assignment.

Three distinctions

A change of labels does not make two orders nonisomorphic. G plus a labeled vector d is stronger input than G plus an unordered degree multiset. A complete finite verification is not an arbitrary-n theorem. These distinctions control every result below.

Reading map

Pages 2-3: directly justified surviving structure. Page 4: scope, open problems and source map. Pages 5-7: unchanged approved v129 degree/scale and maximum-anchor audit, extracted from historical pages 311-313. Pages 8-10: unchanged v130 finite verification, extracted from historical pages 314-316. For the counterexample construction, read 01_COUNTEREXAMPLE/Counterexample_Note.pdf. No physics claim or novelty certification is made.

What survives: direct recovery identities

1. Recover the order from unnormalized K

For every finite poset, $x \leq y$ if and only if $K_{xy} = K_{yy}$. Indeed $|U_x \cap U_y| = |U_y|$ means U_y is contained in U_x ; since y belongs to U_y , $x \leq y$ follows. Conversely $x \leq y$ implies U_y is contained in U_x by transitivity. Thus supplying labeled upset sizes d with G gives $K_{xy} = G_{xy} \sqrt{d_x d_y}$, and then recovers the entire order. The known counterexample shows that an unordered multiset of upset sizes is insufficient in general.

2. Recover the product of upset sizes

In a compatible linear extension Z is upper triangular with diagonal entries one, so $\det(Z) = 1$ and $\det(K) = 1$. Therefore $\det(G) = 1 / \prod_x d_x$. This gives an exact product constraint, not the individual labeled sizes.

3. Recover zero overlap and the number of maxima

$G_{xy} = 0$ exactly when U_x and U_y are disjoint. Every nonempty upset contains a maximal element. Pairwise disjoint upsets therefore require distinct maximal elements; the singleton upsets of all maximal elements attain this bound. The largest set of distinct rows with pairwise zero Gram entries has cardinality equal to the number of maximal elements.

4. Recover comparability components from the zero pattern

Build a graph on row labels, joining distinct x, y when $G_{xy} > 0$. Comparable elements share an upset member, so every comparability edge is a graph edge. Conversely positive overlap supplies z with $x \leq z$ and $y \leq z$, yielding a comparability path through z . The two graphs therefore have exactly the same connected components. This direct proof avoids reliance on the historical principal-minor theorem.

5. Recover the order when a compatible linear extension is supplied

With $M = Z^{-1}$, $G^{-1} = D^{1/2} M^T M D^{1/2}$. The lower-triangular matrix $C = D^{1/2} M^T$ has positive diagonal and satisfies $CC^T = G^{-1}$, so it is the Cholesky factor. Its squared diagonal gives labeled d , then M and Z follow. The supplied compatible ordering is an essential assumption.

What survives: exact realization machinery

6. Test candidate sizes without assuming uniqueness

For nonnegative symmetric input G with diagonal one, let $q_{ij}=G_{ij}$ squared. Given integer candidate sizes d_i in $[1,n]$, define $Z_d(i,j)=1[q_{ij} d_i=d_j]$. Require the poset axioms, row sums equal to d , and $q_{ij} d_i d_j$ equal to the squared row-intersection count for every pair. These tests are necessary and sufficient for realizing G . To incorporate degree data, require the computed comparability-degree multiset to match. Pages 5-7 contain the approved proof and scale constraints.

7. Enumerate candidate maximal layers

In any realization, the true maximal singleton upsets give a maximum set J of pairwise zero-overlap rows. For each x , overlaps with J are either zero or $1/\sqrt{d_x}$, with at least one positive overlap. Conversely, a proposed maximum orthogonal J with those equal positive overlaps and integer reciprocal squared sizes supplies a candidate d . Apply the complete validity test in item 6. Every true realization is covered; a proposed J need not be valid or unique.

8. Check residual closure and maximal attachment

With proposed residual labels R and maximal labels J , write $Z=[[Z_R,A],[0,I]]$. Its Gram product has blocks $Z_R Z_R^T+AA^T$, A , A^T and I . When Z_R is a poset zeta matrix, transitivity of the full relation is equivalent to each attachment column of A being a downset of that residual order. Full validity therefore separates into residual Gram closure and valid maximal attachment, together with the corresponding block equalities. This is a validity reduction, not a confluence theorem.

9. Enumerate every realization when G has positive entries

Every finite realization then has one greatest element. Each possible maximum m fixes all $d_i=1/q_{im}$ and therefore fixes the candidate order by saturation. Check all n anchors. The approved v129 theorem is complete enumeration, not uniqueness. For the original 16-label G exactly two anchors pass, producing P and Q ; their degree multisets differ. The finite audit through $n=8$ is recorded on pages 8-10.

Review boundary

The direct justifications above transport only statements that do not depend on refuted universal uniqueness. They do not globally re-audit every historical conditional branch. The $q=2$ and $L=4$ scoped results and the historical principal-minor formulation remain accessible in the historical evidence, with their existing dependency warnings.

Open problems and evidence map

Current mathematical questions

The central question is whether (G, Δ) , with Δ the unordered comparability-degree multiset, determines a finite poset up to isomorphism. Within the positive-entry class, ask whether two nonisomorphic valid maximum anchors can share Δ . Beyond that class, candidate maximal-layer enumeration remains available but multiple maxima must be handled. Minimal identifying extra information and efficient recovery algorithms are also unresolved; complete finite candidate descriptions do not establish efficiency.

Established failure and scoped success

G alone does not reconstruct every finite poset. The frozen 16-element P, Q pair is an exact counterexample, not a near collision. Their entire positive-entry Gram fiber consists of those two labeled realizations. Adding either degree multiset selects its respective realization in that fiber only. The recent $n \leq 8$ positive-entry audit independently confirms older finite evidence and does not narrow the general larger- n obstruction.

Historical conditional branches

The repaired $q=2$ result has explicit extremal-branch assumptions and upstream dependencies not globally re-audited here. The $L=4$ cone result has exterior-witness and antichain-profile assumptions; general outside-label extension is unsettled. Neither is used to prove the current degree-filter or anchor results. Refuted universal diagonal-orbit rigidity and decorated confluence are historical claims, not surviving general theorems.

Files and reproducibility

01_COUNTEREXAMPLE contains the standalone construction note. 02_REPRODUCE/CORE contains the original fiber and degree-filter sources with a reader runner; execute `python3 02_REPRODUCE/CORE/run_all.py`. 02_REPRODUCE/FINITE_n8 contains the two exhaustive implementations, known-collision control, exact outputs and full independent certificate ledger. 03_OPEN_PROBLEMS_AND_PROVENANCE contains current questions, extraction checks, and the separate preliminary novelty audit. 90_HISTORICAL_EVIDENCE preserves the cumulative PDF and complete frozen ZIP.

Provenance and next step

This is an organization revision of frozen v130, not a new mathematical attack or a new canonical version. Pages 5-10 preserve historical pages 311-316 in order. Older verified identities receive direct explanations on pages 2-3; unreviewed historical branches stay qualified. Novelty and priority remain unresolved. The proposed next attack is structural analysis of degree-matching anchors or a precisely bounded larger- n family. Mathematics remains frozen.

Degree and Scale Import Audit - v129

Decision and scope

The user-authorized import audit approves three precisely scoped mathematical statements for the surviving-results packet: the exact degree filter and scale-sum identity; complete maximum-anchor enumeration for positive-entry G; and the full classification of the original 16-label Gram fiber. Universal G-only injectivity remains false. Universal G-plus-degree injectivity remains unresolved. No Dirac interpretation or search-based universal assertion is imported.

1. Exact degree-filtered candidate description

Assume nonnegative symmetric input G with diagonal one; let $q_{ij}=G_{ij}$ squared, supplied exactly. For any candidate integer vector d in $\{1,\dots,n\}^n$ define $Z_d(i,j)=1[q_{ij} d_i=d_j]$. A candidate realizes G and the supplied unordered degree multiset Delta exactly when Z_d satisfies the poset axioms, its row sums equal d, $q_{ij} d_i d_j$ equals the squared row-intersection count for every pair, and its computed degree multiset equals Delta. Nonnegativity makes squared Gram equality sufficient.

Proof

For a poset, $K_{ij}=|U_i \text{ intersect } U_j|$ and $i \leq j$ exactly when $K_{ij}=d_j$. With $K_{ij}=G_{ij} \sqrt{d_i d_j}$, saturation is precisely $q_{ij} d_i=d_j$. Necessity of all tests follows. Conversely, a passing Z_d is a poset with the required upset sizes, normalized Gram matrix and degree multiset. Thus the finite candidate description is complete for all finite n, without asserting uniqueness or efficiency.

2. Scale sum and bounds

Let γ_i be the computed comparability degree and a_i the downset size. Then $\gamma_i=d_i+a_i-2$. Counting comparable pairs by strict upper sets or strict lower sets gives $\sum d_i=\sum a_i$, so $\sum d_i=n+(\sum \text{Delta})/2$. Bounds are $\max(1,\gamma_i+2-n) \leq d_i \leq \min(n,\gamma_i+1)$. Degrees are unordered input data; their assignment cannot be assumed. The retained product constraint is $\text{product } d_i=1/\det(G)$.

Limit

These are necessary constraints and an exact validity filter, not a uniqueness theorem. The two-element-chain check retains both $d=(1,2)$ and $d=(2,1)$; they are isomorphic by label swap. Reconstruction targets order isomorphism, not one labeled vector.

Maximum-anchor enumeration - approved scoped result

3. Positive-entry completeness theorem

For nonempty finite-poset input G with all entries strictly positive, every realization has exactly one maximum. Two distinct maxima would have disjoint singleton upsets and hence a zero Gram entry. Every finite nonempty poset has a maximum; with a unique maximum, every element lies below it.

Anchor formula

If m is that maximum, its upset size is one and every U_i contains m . Hence $G_{im}=1/\sqrt{d_i}$, so $d_i=1/q_{im}$. Each of the n possible maximum labels therefore fixes every candidate size and, by saturation, the full labeled order. Apply the integer bounds, poset axioms, row-size and exact Gram-closure tests for each anchor. There is at most one valid realization per anchor.

Completeness

Every realization must occur at one of these n anchors. This is a complete enumeration method for this class, not a statement that only one anchor passes. A realization agreeing with G up to permutation can be relabeled into the supplied matrix, so the enumeration also covers the unlabeled question.

Dependencies reviewed

The proof uses only finiteness, singleton maximal upsets, the original principal-upset convention, saturation and exact closure. It uses no universal diagonal-orbit rigidity, decorated confluence, threshold embedding, $q=2$ proof, $L=4$ proof, Dirac result or generator search. Zero-overlap and labeled-scale reconstruction identities survive independently of the refuted uniqueness target.

Exclusions

Inputs with several maximal elements fall outside this positive-entry theorem. The degree multiset need not distinguish all passing anchors at another input. No assertion of minimal identifying data, novelty or new universal injectivity is approved.

Original fiber classification and frozen status

4. Exact original 16-label classification

The original G has only positive entries. All 16 maximum anchors were enumerated: six fail integer upset-size bounds, eight fail reconstructed row sizes, and exactly two satisfy every test. In the original zero-based sector/bitmask labeling, anchor 8 yields exactly Q and anchor 15 yields exactly P, entrywise. The original audited up/down-size certificate proves P and Q nonisomorphic. Therefore this G has exactly two labeled realizations and two order-isomorphism classes.

Degree consequence

Their comparability-degree multisets differ. Adding the P degree multiset leaves exactly P; adding the Q degree multiset leaves exactly Q. This is a complete conclusion for this particular Gram matrix, not an extrapolation from testing only known candidates. Both size-vector sums are 81, so the sum constraint alone does not separate them.

Reproducibility audit

The degree-scale check, full anchor check and independent anchor verifier were rerun from preserved source copies. All three runs succeeded and every retained exact JSON output matched the original bytes. The independent bitset/integer verifier checks every anchor and rejection and directly recovers the two original matrices. No execution failed in this import audit. Prior failed routes remain preserved unchanged.

Firewall and historical limits

B3/B4 family searches remain finite experimental evidence. Graded block lifts, order-complex Dirac spectra and incidence decompositions remain quarantined. No blanket certification of all historical conditional proofs is claimed. Earlier pages remain intact and are dated records; this current notice controls the imported statements. The broader richer-invariant uniqueness question remains unresolved.

Roadmap and artifact map

Completed: v127 counterexample audit; v128 organization; quarantined explorations and bounded tests; v129 scoped import audit. Next proposed, NOT RUN: test or prove whether another positive-entry G can have nonisomorphic valid anchors sharing degree data. Approved statements: 11_SURVIVING_RESULTS/07_APPROVED_IMPORTS_v129. Exact provenance: 09_EXPERIMENTAL_NOTES/Degree_Scale_Maximum_Anchor_Import_Audit_v129. No next attack begins in this freeze.

Positive-entry degree-filter audit - v130

Result and exact scope

The complete positive-entry class through eight elements has no nonisomorphic Gram realizations. Every valid maximum anchor is isomorphic to the generating order, so degree data are not needed within this finite scope. Universal G-only injectivity remains false by the frozen 16-element counterexample. Universal G-plus-degree injectivity remains unresolved.

Why the enumeration is complete

A finite realization with all Gram entries positive has a unique maximal element, hence a greatest element. Conversely a greatest element makes every upset overlap positive. Choose a compatible linear extension ending at the greatest element; its deletion leaves an arbitrary poset on $n-1$ labels. Enumerating every transitive upper-triangular relation there and appending the greatest element covers every isomorphism class in the target class, with repetitions.

Every possible realization is checked

Set $q_{ij}=G_{ij}$ squared. For each possible maximum m , the unique candidate sizes are $d_i=1/q_{im}$. Require integer values in $[1,n]$, define $Z_{ij}=1[q_{ij} d_i=d_j]$, then check poset axioms, row sizes and exact squared Gram closure. Positive entries ensure that squared equality implies equality of G . Every realization occurs at one of these anchors by the approved v129 completeness theorem.

Interpretation

This independently confirms the positive-entry portion of the historical $n \leq 8$ evidence. It is not a new general theorem, a novelty claim or a narrowing of the larger- n obstruction. Multiple labeled anchors are not structural collisions when an exact order isomorphism relates them.

Exact results and independent verification

Counts

Counts below are orders in compatible labelings, not unlabeled-poset counts or distinct Gram matrices. In total: 101,660 source orders, 807,571 anchor checks, 103,562 valid anchors, and zero nonisomorphic degree-matching candidates.

n	Source orders	Anchors	Valid anchors	Collisions
1	1	1	1	0
2	1	2	2	0
3	2	6	2	0
4	7	28	10	0
5	40	200	45	0
6	357	2,142	396	0
7	4,824	33,768	4,992	0
8	96,428	771,424	98,114	0

Independent implementation

The primary script uses bitsets and integer cross-multiplication. A second script independently generates orders by adding a final vertex with every possible predecessor ideal; it uses sets and rational arithmetic. Both agree on every source count, valid-anchor histogram and rejection category. Every valid order has an explicitly checked isomorphism certificate.

Known-collision control

The unchanged 16-element fixture returns exactly anchors 8 and 15, producing Q and P. The checker detects their nonisomorphism and differing comparability-degree multisets. The original G-only counterexample remains exact and valid.

Timing and execution history

Primary runtime: about 16.24 seconds. Independent runtime: about 40.83 seconds. These are observed timings, not performance guarantees. No execution failed in this attack. Scripts, actual stdout/stderr, exact outputs, the unchanged control input, and every independent source/anchor/candidate/isomorphism record are retained.

Frozen status, dependencies and roadmap

Completed and frozen

v127: counterexample and proof-dependency audit. v128: surviving-results organization. v129: scoped degree/scale and maximum-anchor import audit. v130: this complete finite positive-entry verification through eight elements. Earlier pages and records remain preserved.

Mathematical dependencies

Uses only principal-upset conventions, the approved maximum-anchor completeness reduction, saturation and exact closure. No universal confluence or diagonal-orbit uniqueness claim is used. Conditional $q=2$ and $L=4$ results are not dependencies. Dirac explorations and bounded B3/B4 searches remain quarantined or experimental.

Current frontier

General G-plus-degree injectivity is unresolved. Other positive-entry inputs above eight elements may have nonisomorphic degree-matching valid anchors; this test neither excludes nor constructs them. Inputs with several maximal elements are also outside the present test.

Next proposed, not run

Analyze the structural constraints on two valid maximum anchors with matching degree multisets, or select a precisely bounded larger- n family capable of a new collision. Repeating already covered small- n searches should not be represented as new general progress. No next attack begins during this freeze.

Record map

Full report and executed sources:

09_EXPERIMENTAL_NOTES/Positive_Entry_Degree_Filter_Exhaustive_n8_v130. Current reader entry: 00_START_HERE/READABILITY_GUIDE/CURRENT_STATE_v130.md. Independent certificates: independent_ledger.jsonl.gz. This is an experimental finite result recorded in the canonical history, without promotion to an all- n theorem.